AI-Based Smart Driving License System for Sri Lanka

A digital, intelligent approach to traffic enforcement and license management

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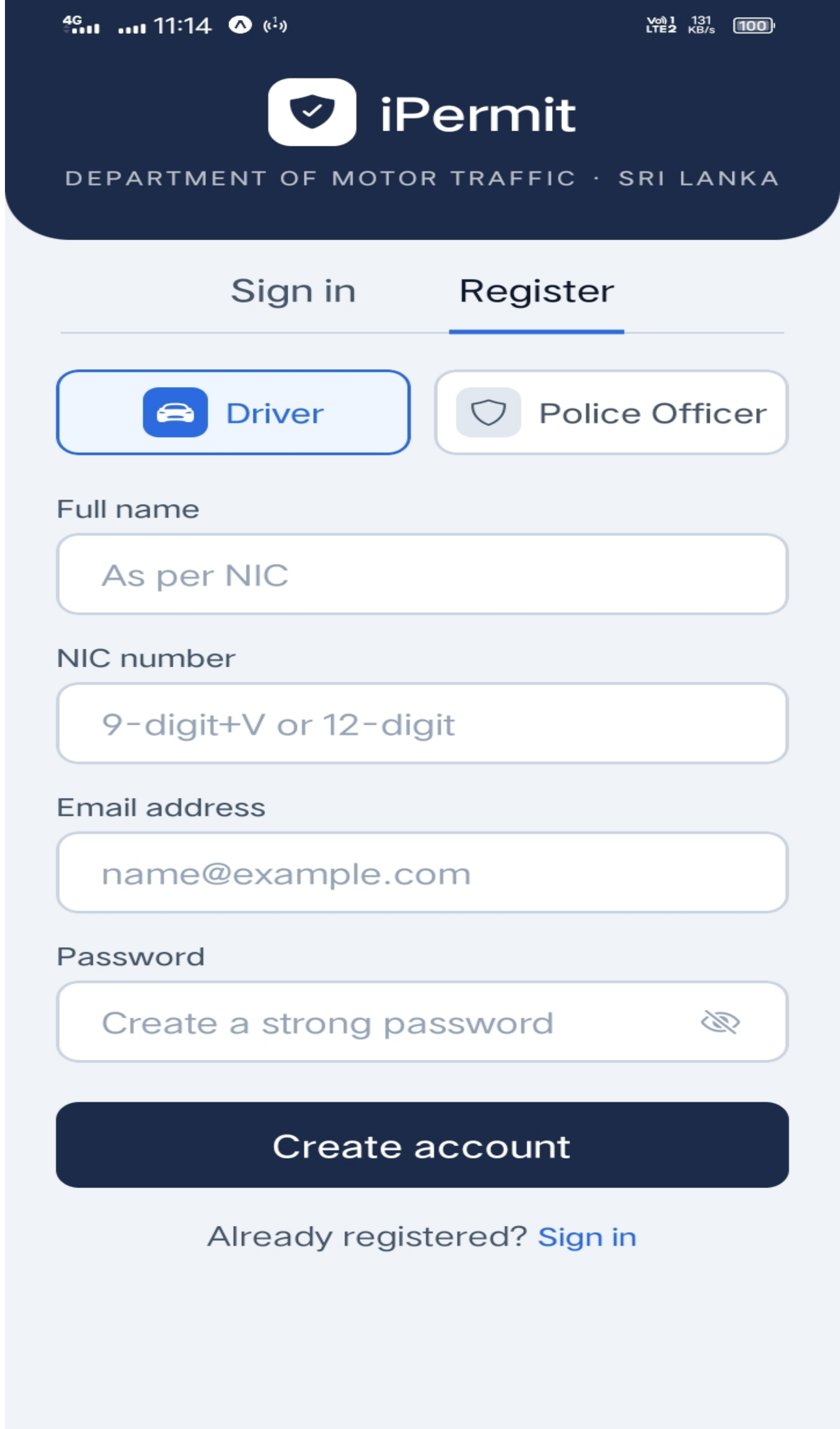
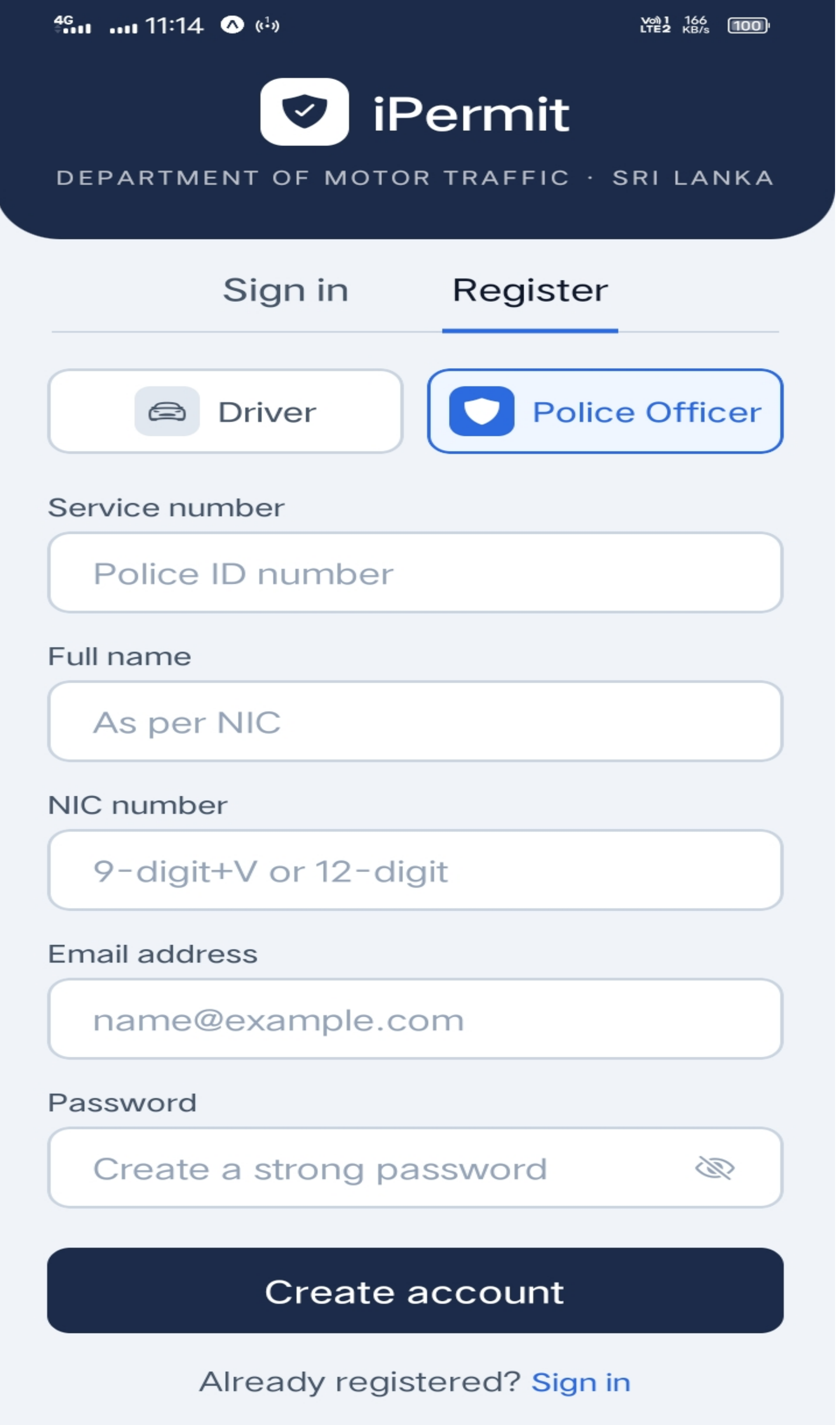




The Problem Statement

Traffic law enforcement in Sri Lanka remains largely manual, creating systemic inefficiencies in license verification and fine management.

Fragmented Systems

Lack of real-time communication between police and licensing authorities.

Identity Fraud

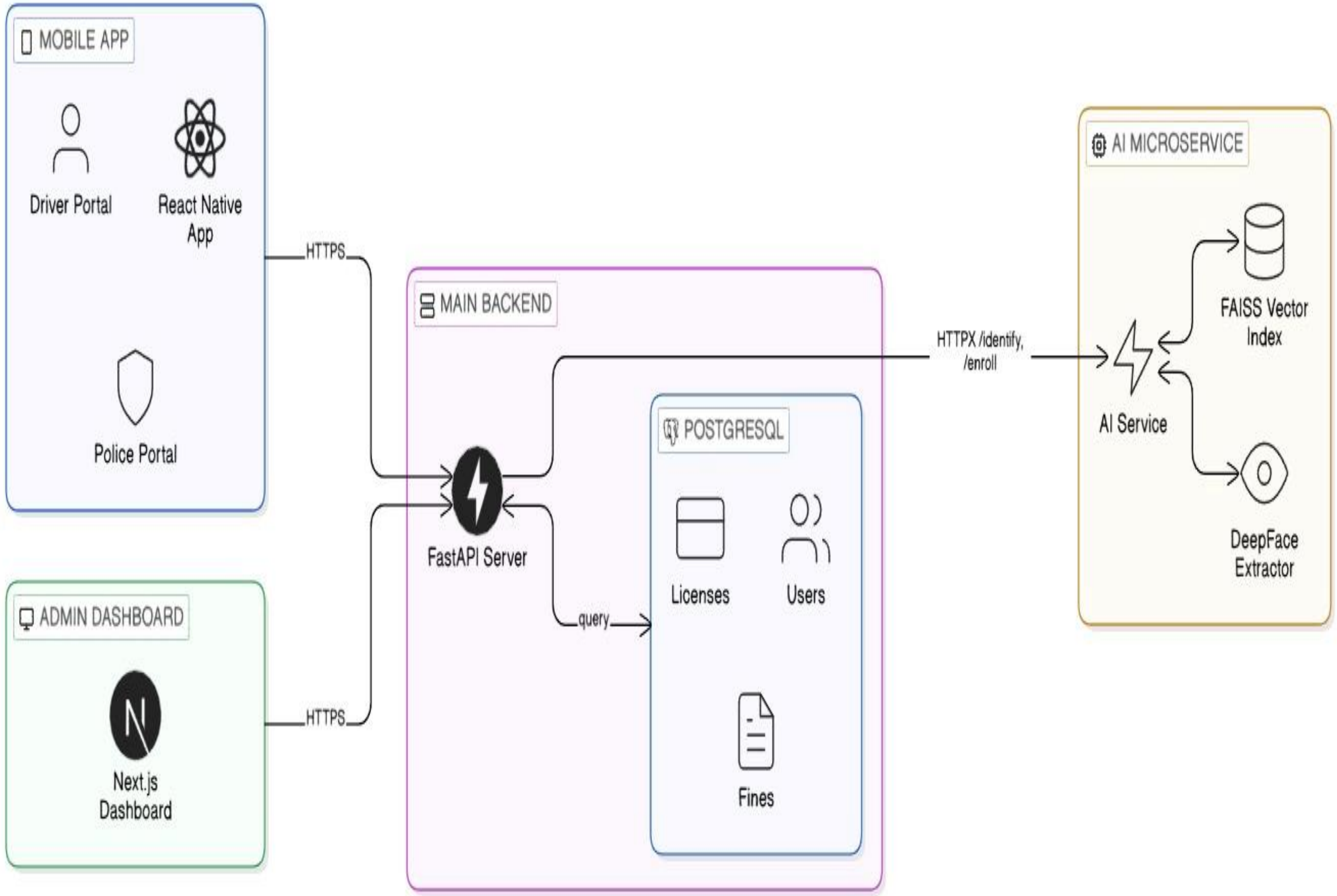
Difficulty in verifying physical licenses at the roadside.

Manual Overhead

Inefficient paper-based applications and fine payment processes.

AI is Transforming Traffic Enforcement

Growing road accidents highlight the urgent need for digital, secure, and modernized systems. This project proposes a comprehensive AI-powered solution.



→ **Digital License Management**
Web & Mobile platform for drivers and officers

→ **Facial Recognition**
Real-time identity verification

→ **Point-Based Violation System**
Behavior tracking and penalty management

→ **Fine Management**
Online payment and violation tracking

→ **Danger Area Detection**
Accident-prone zone identification

→ **White-Line Violation Detection**
Automated lane rule enforcemen

System Architecture

1 **Core Backend**

FastAPI orchestrating auth, fine issuance, and application workflows.

2 **AI Microservice**

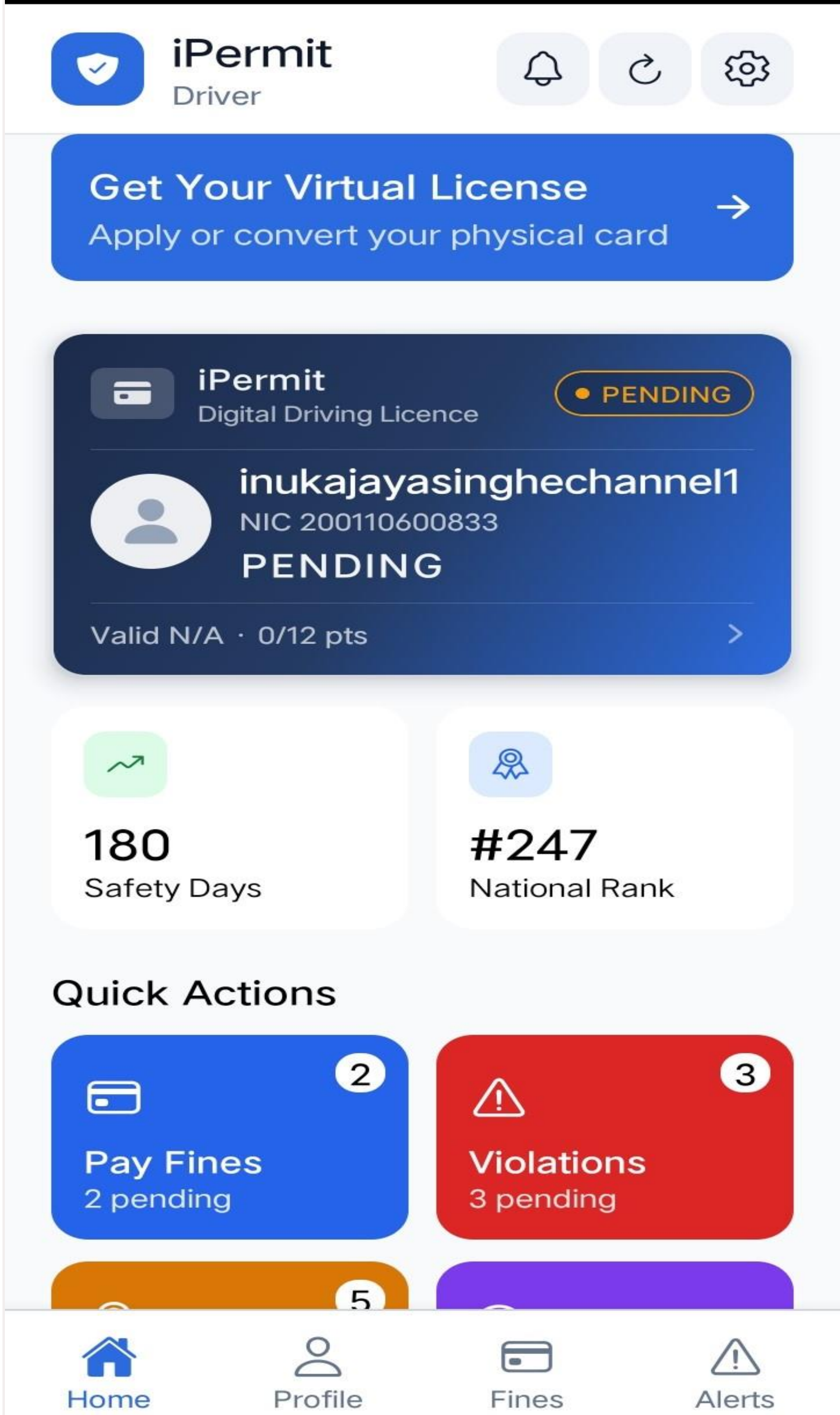
Standalone module for face enrollment and high-speed identification.

3 **Mobile Client**

React Native/Expo app with specialized portals for Drivers and Police.

4 **Web Portal**

Next.js dashboard for administrative oversight.



Technical Stack

1

Frontend

React Native (Driver/Police App), Next.js (Admin).

2

Backend

Python (FastAPI), PostgreSQL (Database).

3

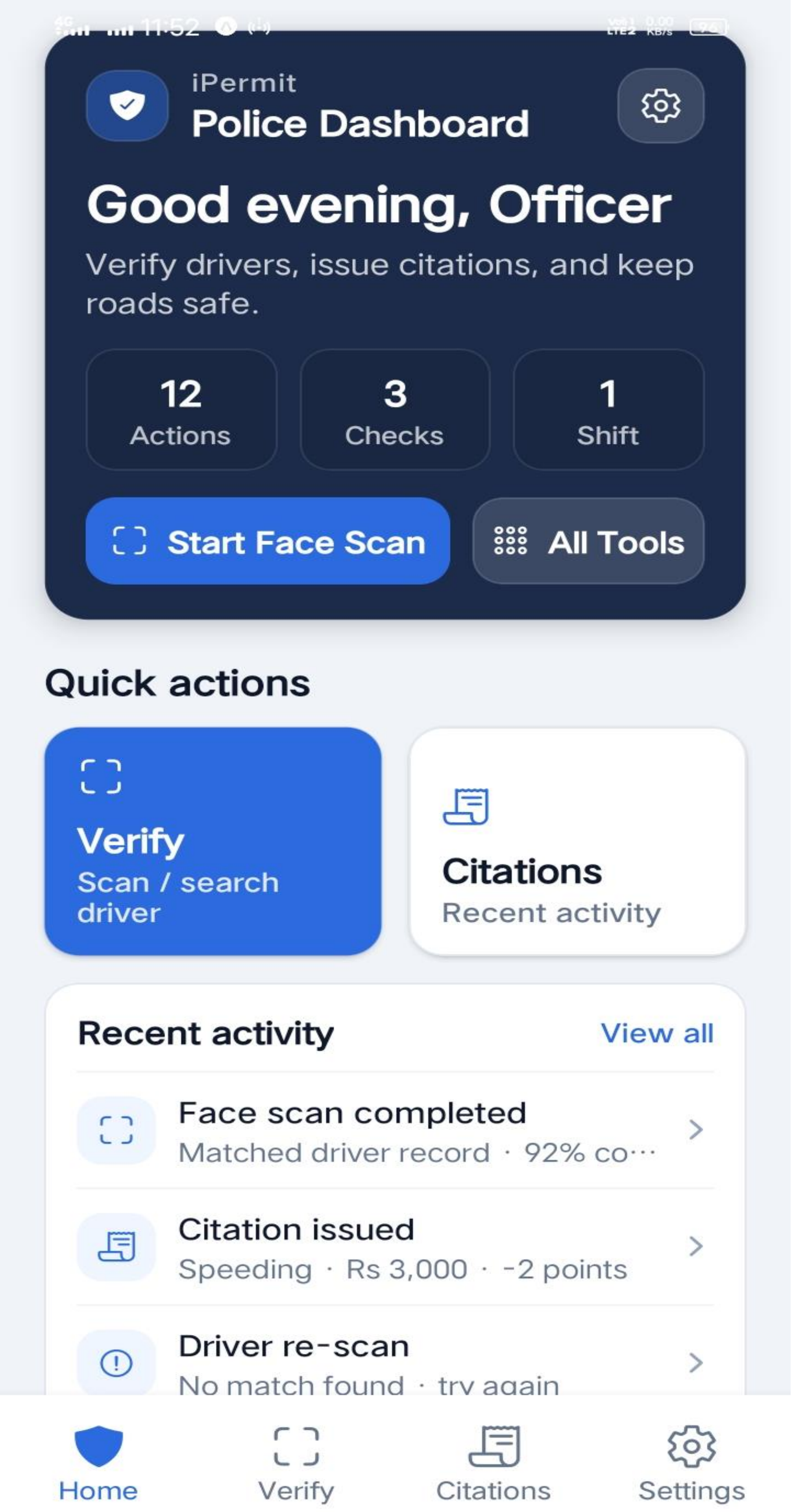
AI

DeepFace (Facenet), FAISS (Vector Index), RetinaFace (Detection).

4

DevOps

Docker-compose for microservice orchestration.



AI Facial Recognition

Biometric identity verification using deep learning models enables fast, contactless authentication for drivers and traffic officers.

Preprocessing

CLAHE for lighting normalization and RetinaFace for eye-landmark alignment.

Feature Extraction

Generating a **128-dimensional embedding** using the Facenet model.

Matching Logic

Comparing normalized vectors using **FAISS Squared L2 distance** with a Threshold of **0.75**

Driver Portal: Digital License Management

Home

View virtual license status and real-time point balance.

Documents

Uploading NIC, NTMI Medical, and Birth Certificates for digital conversion.

Fines

View citation history and proceed to mock payment checkout.

Fine Management System

Fine Details

Complete violation records with dates, locations, and descriptions accessible to drivers and officers.

Total Fines

Aggregated fine summaries showing outstanding and paid amounts at a glance.

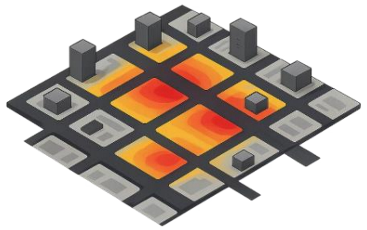
Payment Methods

Multiple online payment options for convenient, cashless fine settlement.

Point Deduction

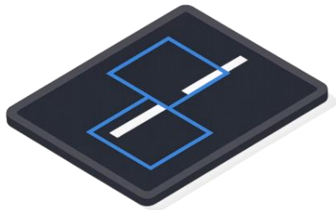
Automatic point tracking linked to each fine, updating the driver's violation score in real time.

Future Plans: Advanced AI Modules



Danger Area Detection

Identifies accident-prone zones using historical data. ML models: ***Random Forest*** *&* ***XGBoost****.*



White-Line Violation Detection

Detects lane and stop-line violations via ***Computer Vision, YOLO*** *& Deep Learning with real-time camera tracking.*

Number Plate Recognition (NPR)

Automates vehicle identification using ***CNN*** *and* ***OCR*** *for violation tracking, reducing manual workload.*

Mobile & Web Applications

A unified digital platform providing easy access for both drivers and traffic enforcement officers.

License Verification

Face recognition for instant identity confirmation

Violation Alerts

Real-time notifications for traffic infractions

Online Payments

Seamless digital fine settlement

License Services

Convert, renew, or apply for a new digital license

Conclusion

AI-powered traffic management is a future-ready solution for Sri Lanka — with proper planning, it can transform urban mobility.

Safety

Real-time monitoring reduces accidents and improves emergency response.

Efficiency

Adaptive signal control minimizes congestion and travel time.

Transparency

Open data and clear decision-making build public trust.

Thank You

We appreciate your time and attention. Your questions and feedback are welcome.

Questions?

Open the floor for discussion and inquiries.

Next Steps

Explore pilot opportunities and stakeholder engagement for Sri Lanka's smart traffic future.